Ex: 07
10.9.24

SLIDING WINDOW PROTOCOL

AIM: Write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL.

CODE:

Sender.py:

```
import time
import os
sender_file = "sender_Buffer.txt"
receiver_file= "Receiver_Buffer.txt"
win_size = int(input(" Enter window size:"))
message = input("Enter text message")
frames = [ ]
for i in range (len (message)):
        frames.append((i+1, message[i]))
base=0
while base < len (frames):
    end = min (base + window_size, len (frames))
    with open(sender_file, "w") as file:
        for i in range (base, end):
            frame_no, data = frames[i]
            file.write(f" Frame No: {frame_no},
                        DATA : {data} \n")
    print ("sending frames")
    for i in range(base, end):
        frame_no, data = frames [i]
        print(f" Frame No : {frame_no}, DATA: {data}")
    print(" \n waiting - ack")
    time. sleep (3)
    while not os.path.exists (receiver_file):
        time.sleep(1)
```

```
with open (receiver_file, "r"
elif response.startswith ("NACK"):
    nack_no = int(response.split(":")[1])
    print("NACK received for frame:", nack_no)
    print("Resending from frame:", nack_no)
    base = nack_no - 1
with open (receiver_file, "w") as file:
    file.write("")
print("In All frames sent successfully")
```

Receiver.py:

```
import time
import os
sender_file = "sender_Buffer.txt"
receiver_file = "Receiver_Buffer.txt"
expected_frame = 1
while True:
    if not os.path.exists(sender_file):
        time.sleep(1)
        continue
    with open (sender_file, "r") as file:
        frames = file.readlines()
    if not frames:
        time.sleep(1)
        continue
    print("In Frames received:")
    error = False
```

```
if not error:
    ack_no = expected_frame - 1
    with open(receiver_file, "w") as file:
        file.write(f"ACK {ack_no}")
    print("Ack sent:", ack_no)
time.sleep(1)
```

RESULT:

Hence the code for sliding window was successfully executed.